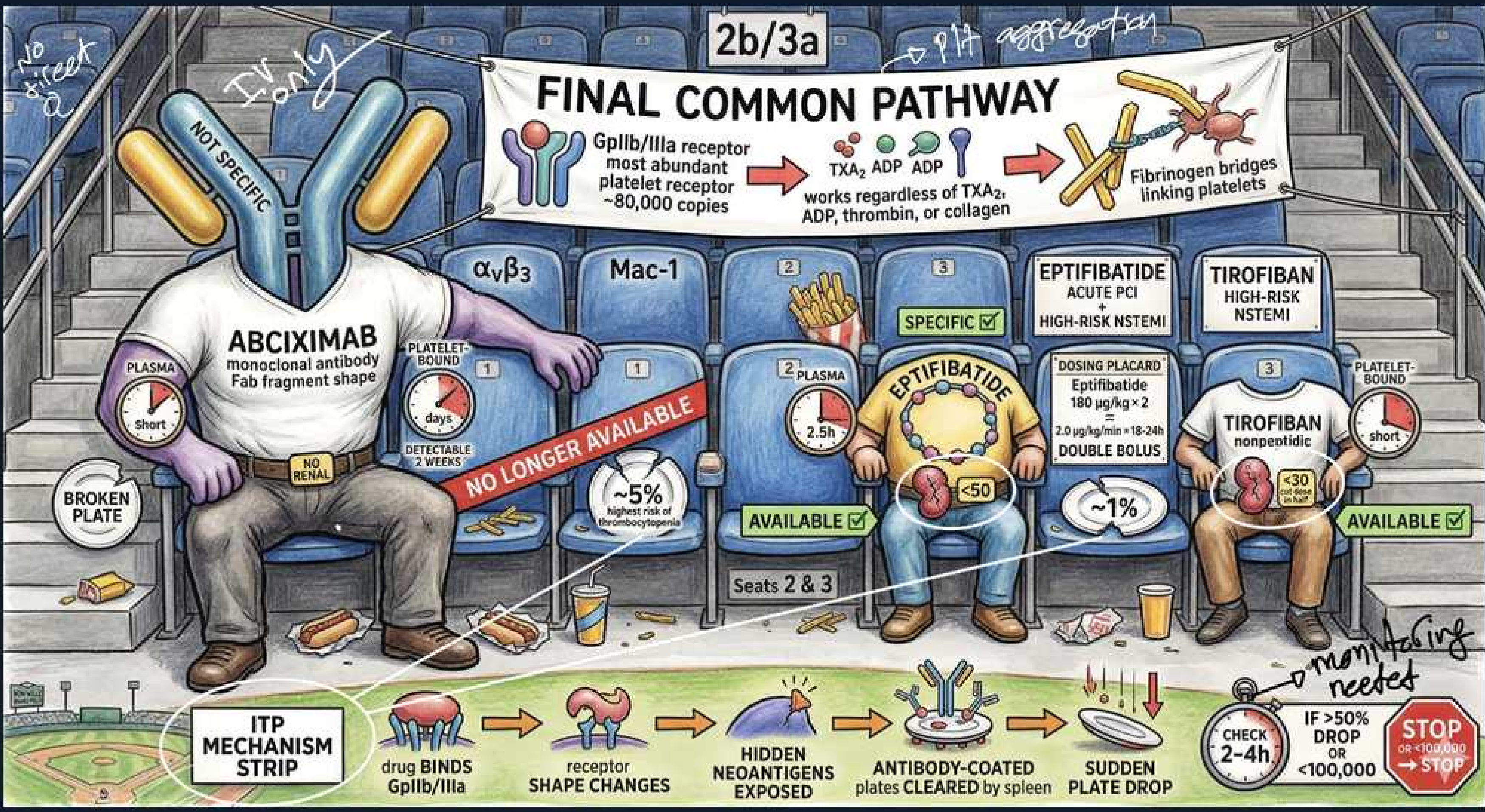
GP IIb/IIIa Inhibitors — Abciximab, Eptifibatide, Tirofiban



1. Final common pathway banner

WHAT YOU SEE

Stadium banner 'FINAL COMMON PATHWAY' with '2b/3a' and 'plt aggregation' written above.

WHAT IT ENCODES

Glycoprotein IIb/IIIa is the final common pathway of platelet aggregation. Once activated it binds fibrinogen, which cross-bridges adjacent platelets. Blocking it stops aggregation regardless of the upstream trigger (TXA2, ADP, thrombin, or collagen).

BOARD TRAP

GP IIb/IIIa blockade works downstream of ALL agonists — aspirin and clopidogrel block only one pathway each, so GP IIb/IIIa is the most complete aggregation block.

---

2. GP IIb/IIIa receptor — most abundant

WHAT YOU SEE

'GpIIb/IIIa receptor — most abundant platelet receptor ~80,000 copies'.

WHAT IT ENCODES

GP IIb/IIIa (integrin αIIbβ3) is the most abundant platelet surface receptor (~80,000 per platelet). When platelets activate it changes shape to bind fibrinogen, the molecular bridge that links platelets into an aggregate.

BOARD TRAP

Fibrinogen — not fibrin or vWF — is the primary ligand bridging platelets via GP IIb/IIIa in the final common pathway.

---

3. Fibrinogen bridges

WHAT YOU SEE

'Fibrinogen bridges linking platelets' graphic, TXA2/ADP/ADP arrows.

WHAT IT ENCODES

Fibrinogen binds activated GP IIb/IIIa on two platelets simultaneously, forming the cross-bridge that creates the platelet plug. GP IIb/IIIa inhibitors occupy this site so fibrinogen cannot bridge.

BOARD TRAP

All three agents act 'regardless of TXA2, ADP, thrombin, or collagen' — that universality is the GP IIb/IIIa class hallmark.

---

4. Abciximab — monoclonal Fab

WHAT YOU SEE

Large 'ABCIXIMAB' figure: 'monoclonal antibody Fab fragment shape', αvβ3 / Mac-1.

WHAT IT ENCODES

Abciximab is a chimeric monoclonal antibody Fab fragment. It binds GP IIb/IIIa with high affinity but LOW specificity — it also blocks αvβ3 (vitronectin) and Mac-1 receptors. Long biologic half-life (receptor-bound ~2 weeks) despite short plasma half-life.

BOARD TRAP

Abciximab is NOT specific — it cross-reacts with αvβ3 and Mac-1, unlike the specific small-molecule agents eptifibatide and tirofiban.

---

5. Abciximab short plasma / long biologic

WHAT YOU SEE

'PLASMA short' but 'reversible 2 weeks' / receptor-bound 2 weeks.

WHAT IT ENCODES

Abciximab clears from plasma in minutes but stays receptor-bound for up to ~2 weeks, so platelet inhibition outlasts the infusion. Reversal of bleeding requires platelet transfusion.

BOARD TRAP

Abciximab has a SHORT plasma half-life but a LONG biologic effect (receptor-bound weeks) — the opposite profile of the small molecules.

---

6. Abciximab thrombocytopenia ~5%

WHAT YOU SEE

'~5%' tag — drop in platelets.

WHAT IT ENCODES

Abciximab causes the highest rate of acute drug-induced low platelet counts among the class (~5%, sometimes profound within hours) because it is an antibody. Check counts 2–4 hours after starting.

BOARD TRAP

Highest thrombocytopenia risk = abciximab (it's an antibody); eptifibatide ~1%, tirofiban low — antibody nature drives the immune drop.

---

7. Abciximab no longer available

WHAT YOU SEE

Red banner 'NO LONGER AVAILABLE' across the abciximab seat; 'broken plate'.

WHAT IT ENCODES

Abciximab has been discontinued/withdrawn from many markets — a historical/board-trivia agent now. The small molecules eptifibatide and tirofiban remain available and in use.

BOARD TRAP

Abciximab is the antibody you must KNOW for exams but is no longer commercially available — clinically you reach for eptifibatide/tirofiban.

---

8. Eptifibatide — cyclic heptapeptide

WHAT YOU SEE

'EPTIFIBATIDE' figure: 'SPECIFIC', 'ACUTE PCI / HIGH-RISK NSTEMI', plasma 2.5 h.

WHAT IT ENCODES

Eptifibatide is a small cyclic heptapeptide (KGD-based) that SPECIFICALLY and reversibly blocks GP IIb/IIIa. Plasma half-life ~2.5 h; used in acute PCI and high-risk NSTEMI. Renally cleared.

BOARD TRAP

Eptifibatide is specific (GP IIb/IIIa only) and reversible — unlike abciximab; it is renally cleared so dose-adjust in renal impairment.

---

9. Eptifibatide double bolus + infusion

WHAT YOU SEE

Dosing placard: 'Eptifibatide 180 µg/kg x2 DOUBLE BOLUS', '2.0 µg/kg/min infusion'.

WHAT IT ENCODES

Eptifibatide dosing: two 180 µg/kg boluses 10 min apart, then 2.0 µg/kg/min infusion (reduce to 1.0 in renal impairment). The double bolus distinguishes its regimen.

BOARD TRAP

Double 180 µg/kg bolus = eptifibatide. Contraindicated in dialysis-dependent renal failure; reduce infusion when CrCl is low.

10. Eptifibatide thrombocytopenia ~1%

WHAT YOU SEE

'~1%' low-platelet tag at the eptifibatide seat.

WHAT IT ENCODES

Eptifibatide causes thrombocytopenia in roughly 1% — much lower than abciximab because it is a small peptide, not an antibody.

BOARD TRAP

~1% thrombocytopenia distinguishes eptifibatide from abciximab's ~5% — small molecules are less immunogenic.

11. Tirofiban — nonpeptide

WHAT YOU SEE

'TIROFIBAN' figure: 'HIGH-RISK NSTEMI', 'nonpeptide', plasma short, 'PLATELET-BOUND short'.

WHAT IT ENCODES

Tirofiban is a small nonpeptide tyrosine-derivative that reversibly and specifically blocks GP IIb/IIIa. Short plasma half-life (~2 h), used in high-risk NSTEMI/ACS and PCI. Renally cleared — reduce dose in renal impairment.

BOARD TRAP

Tirofiban is the NONpeptide; eptifibatide is the cyclic peptide — both specific, reversible, renally cleared small molecules.

12. All three available except abciximab

WHAT YOU SEE

Green 'AVAILABLE' checks on eptifibatide and tirofiban seats.

WHAT IT ENCODES

Both small-molecule GP IIb/IIIa inhibitors (eptifibatide, tirofiban) remain available for high-risk ACS/PCI; abciximab is discontinued. All three are reserved for high-thrombotic-risk PCI and are stopped before surgery.

BOARD TRAP

Class indication is high-risk NSTEMI/ACS during PCI — not chronic oral therapy; these are IV, in-hospital agents only.

13. Mechanism strip — receptor shape change

WHAT YOU SEE

Bottom strip: 'drug BINDS GpIIb/IIIa → receptor SHAPE CHANGES → hidden neoantigens exposed → antibody-coated → sudden plate drop'.

WHAT IT ENCODES

Mechanism of GP IIb/IIIa-induced thrombocytopenia: drug binding triggers a receptor conformational change exposing hidden neoepitopes, which preformed antibodies recognize; antibody-coated platelets are cleared by the spleen, causing sudden severe thrombocytopenia.

BOARD TRAP

This acute thrombocytopenia is a conformational-neoepitope/antibody mechanism — distinct from heparin-induced low platelets; treat by stopping the drug.

14. Check counts 2–4h

WHAT YOU SEE

Red note: 'CHECK 2–4h' and 'IF >50% DROP or <100,000 → STOP'.

WHAT IT ENCODES

Monitor platelet count 2–4 hours after starting a GP IIb/IIIa inhibitor (and again at ~24 h). Stop the drug if platelets fall >50% from baseline or below 100,000 — abrupt severe drops can occur within hours, especially with abciximab.

BOARD TRAP

Stop immediately for >50% platelet drop or <100,000 — the 2–4 h early check catches abciximab's rapid antibody-mediated thrombocytopenia.

15. Contraindications — bleeding risk

WHAT YOU SEE

Stadium debris / bleeding theme; main adverse effect is hemorrhage.

WHAT IT ENCODES

As the most potent antiplatelet class, GP IIb/IIIa inhibitors' chief risk is major bleeding. Avoid with active bleeding, recent stroke, severe uncontrolled hypertension, or thrombocytopenia; reverse abciximab bleeding with platelet transfusion.

BOARD TRAP

Bleeding is the dominant class toxicity; for abciximab the antibody stays bound so platelet transfusion is the rescue, whereas small-molecule effect simply wears off.